

Lesson 1: Computer Hardware and Software

Learning objectives

- Identify the main parts of a computer system.
- Explain the difference between hardware and software.

1.1 Hardware

Hardware means the physical parts of a computer that you can touch. It includes input devices (keyboard, mouse, scanner), output devices (monitor, printer, speakers), the processor (CPU), memory (RAM) and storage (hard disk or SSD).

1.2 Software

Software is the set of programs that tell the hardware what to do. **System software** such as an operating system (Windows, Linux, Android) manages the computer. **Application software** such as word processors, browsers and games helps users do specific tasks.

1.3 The input-process-output cycle

Every computer follows the same cycle: data is entered through an input device, the CPU processes it using programs held in RAM, and the result is shown on an output device or saved to storage.

Part	Type	Role
CPU	Hardware	Processes instructions
RAM	Hardware	Temporary working memory
SSD / Hard disk	Hardware	Long-term storage
Operating system	System software	Manages hardware and programs
Web browser	Application software	Displays web pages

Key points

- Hardware is physical; software is programs.
- RAM loses its contents when power is off.
- Storage keeps data permanently.

Review questions

- 1 Give two examples each of input and output devices.
- 2 What is the role of an operating system?
- 3 State one difference between RAM and storage.

Lesson 2: Networks and the Internet

Learning objectives

- Define a network and describe LAN and WAN.
- Explain what an IP address and a router do.

2.1 What is a network?

A network is two or more devices connected so that they can share data and resources such as files, printers and an internet connection.

2.2 Types of network

A **LAN** (Local Area Network) covers a small area such as a school lab or home. A **WAN** (Wide Area Network) covers cities or countries. The **Internet** is a global network of networks.

2.3 Key devices and ideas

A **router** connects networks together and directs data to the right place. A **switch** connects devices inside one LAN. Each device has an **IP address**, a number that identifies it on the network, similar to a house address. A **URL** such as www.example.com is a human-friendly name that DNS converts into an IP address.

Term	Meaning
LAN	Network in a small area, for example a school
WAN	Network covering a large area
Router	Joins networks and directs traffic
IP address	Unique number identifying a device
DNS	Turns website names into IP addresses

Key points

- The Internet is a network of networks.
- Routers join networks; switches connect devices in a LAN.
- DNS works like a phone book for websites.

Review questions

- 1 Give one example of a LAN and one of a WAN.
- 2 What does a router do?
- 3 Why do we need DNS?

Lesson 3: Staying Safe Online

Learning objectives

- Recognise common online threats.
- Apply simple habits that protect personal data.

3.1 Common threats

Malware is harmful software such as viruses and spyware. **Phishing** uses fake emails or messages to trick people into giving passwords or bank details. **Identity theft** happens when someone uses your personal information without permission.

3.2 Good habits

Use long, unique passwords and turn on two-factor authentication.

Keep the operating system and antivirus software up to date.

Do not click links or open attachments from unknown senders.

Think before sharing personal details or photos on social media.

Back up important files regularly.

Threat	What it does	Protection
Virus / malware	Damages or spies on your device	Antivirus, updates
Phishing	Tricks you into giving details	Check sender, do not click
Weak password	Easy to guess or crack	Long unique passwords

Key points

- Never share your password.
- Be suspicious of urgent messages asking for details.
- Backups protect against data loss.

Review questions

- 1 What is phishing?
- 2 List three features of a strong password.
- 3 Why should software be kept up to date?